

Egor Gikalo

646-282-6110 | esgikalo@gmail.com | [LinkedIn](#) | [Website](#)

EDUCATION

New York University

New York, NY

Master of Science in Data Science

Sep 2025 – May 2027

- **Coursework:** Computer Vision, Natural Language Processing, Machine Learning, Optimization and Computational Linear Algebra, Introduction to Data Science, Big Data

Lomonosov Moscow State University

Moscow, Russia

Bachelor of Science in Applied Mathematics and Computer Science

Sep 2021 – Jun 2025

- **Awards:** State Academic Scholarship, Gold Medal in Annual Thesis Competition 2025
- **Activities:** MSU Admission Ambassador, Lead Event Coordinator

WORK EXPERIENCE

New York University CILVR

New York, NY

Research Assistant (advised by prof. Saining Xie)

Jan 2026 – present

- **Developed** the first multiplayer video world model using **Diffusion Transformers** and flow matching.
- **Introduced** a Checkpointed Self Forcing technique that significantly reduced memory usage and improved visual consistency during long-horizon autoregressive training. Implemented in **JAX**.
- **Built** a scalable multiplayer data collection framework (SolarisEngine) integrating Mineflayer and **Docker** to synchronize multi-agent actions and visual observations, successfully curating a 12.6M-frame annotated Minecraft training dataset.

Russian Academy of Sciences

Moscow, Russia

Student Researcher

Jan 2023 – Jul 2025

- **Designed** a novel graph neural network-based method for entity resolution in knowledge graphs using link prediction techniques, achieving a **14%** improvement in accuracy over baseline methods using **PyTorch**.
- **Built** a dataset conversion module for RAS's graph neural networks framework enabling interoperability between **NetworkX** and **PyTorch**, streamlining graph preprocessing and expanding framework flexibility.

TheStage AI

Remote

Machine Learning Engineer Intern

Jun 2024 – Sep 2024

- **Integrated** TheStage AI's internal quantization library with EleutherAI's "lm-evaluation-harness" by developing a custom model adapter, enabling standardized benchmarking of optimized/quantized LLMs (beyond Hugging Face models), accelerating research scientists' ability to run on-the-fly evaluations.

PROJECTS

CVJutsu | Python, OpenCV, MediaPipe, scikit-learn, PyQt6

- Built a real-time desktop computer vision app that classifies 12 Naruto hand seals from live webcam input using MediaPipe hand landmark extraction and a Random Forest classifier trained on custom-collected gesture data.
- Implemented a multi-stage ML pipeline with data augmentation, confidence thresholding, and temporal sequence tracking to recognize 7 hand seal combinations, with a PyQt6 GUI overlay rendering animated visual effects on detection.

SKILLS

Languages: Python, SQL (PostgreSQL), C, C++

Tools & Technologies: Git, CI/CD, Docker, Jupyter Notebook, vim, Bash, AWS, Apache Spark

Libraries: pytorch, jax, pandas, numpy, matplotlib, scipy, scikit-learn, nltk, transformers, diffusers